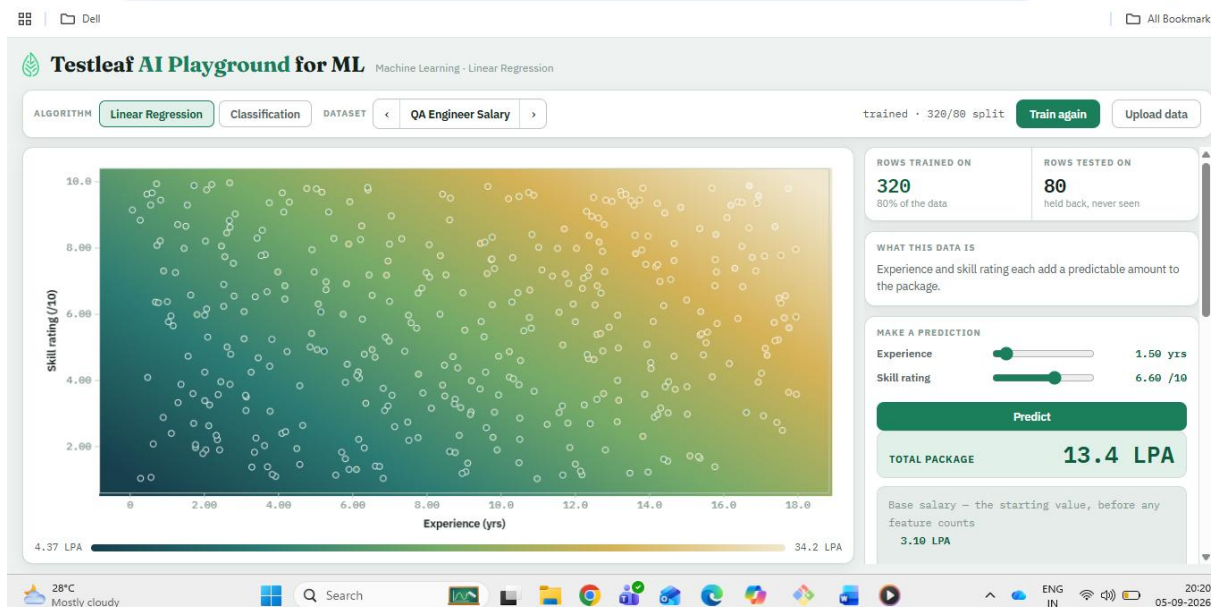
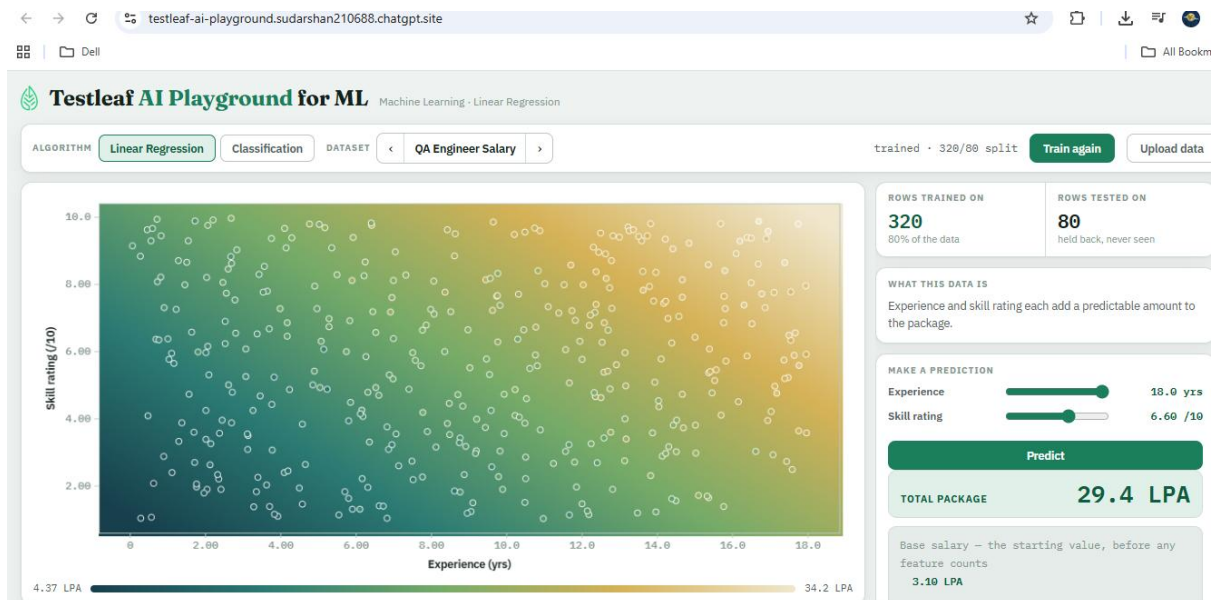


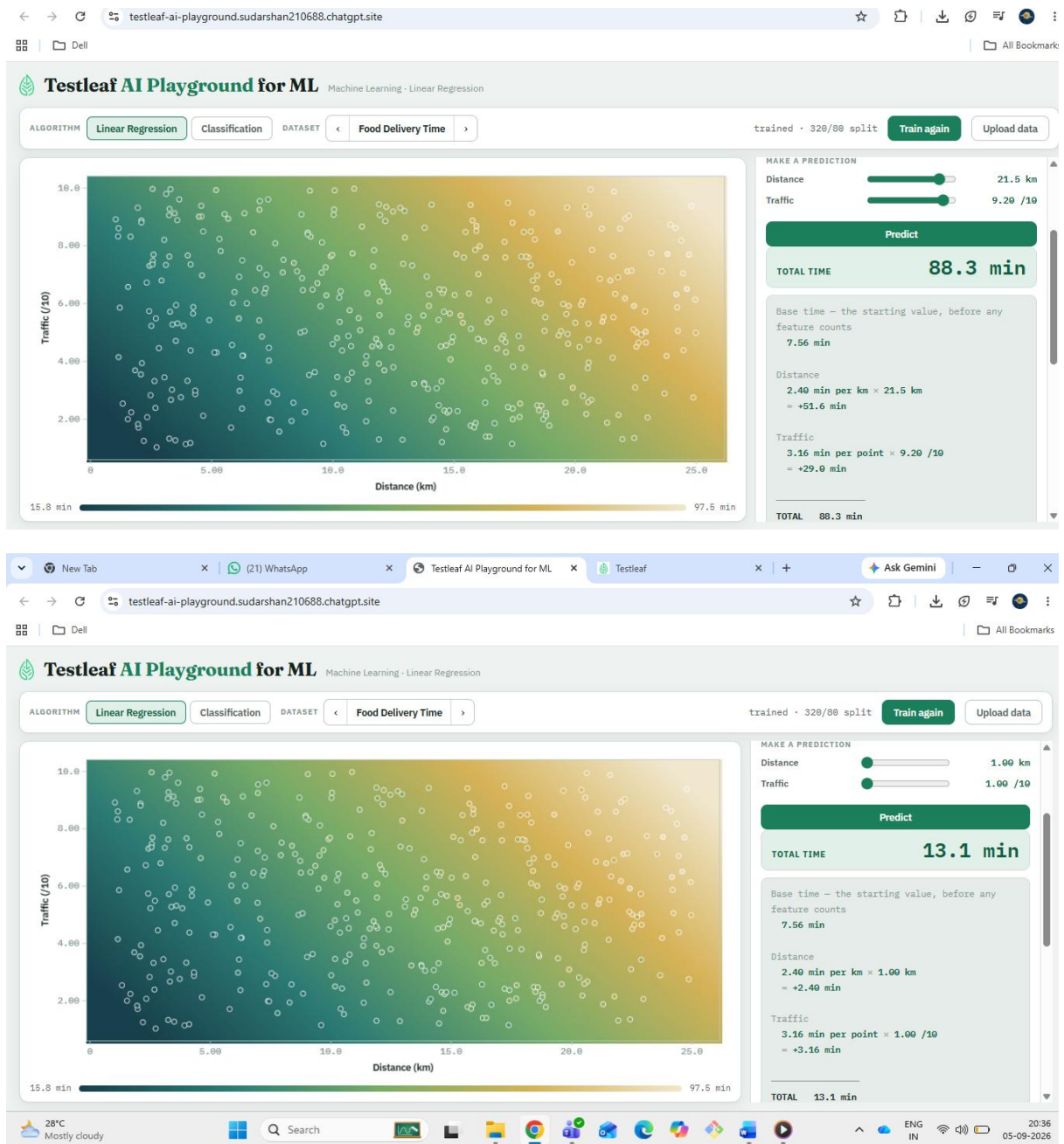
week01-day-1-HomeAssignment

Linear regression

QA engineer salary [varying as per skill and years of experience

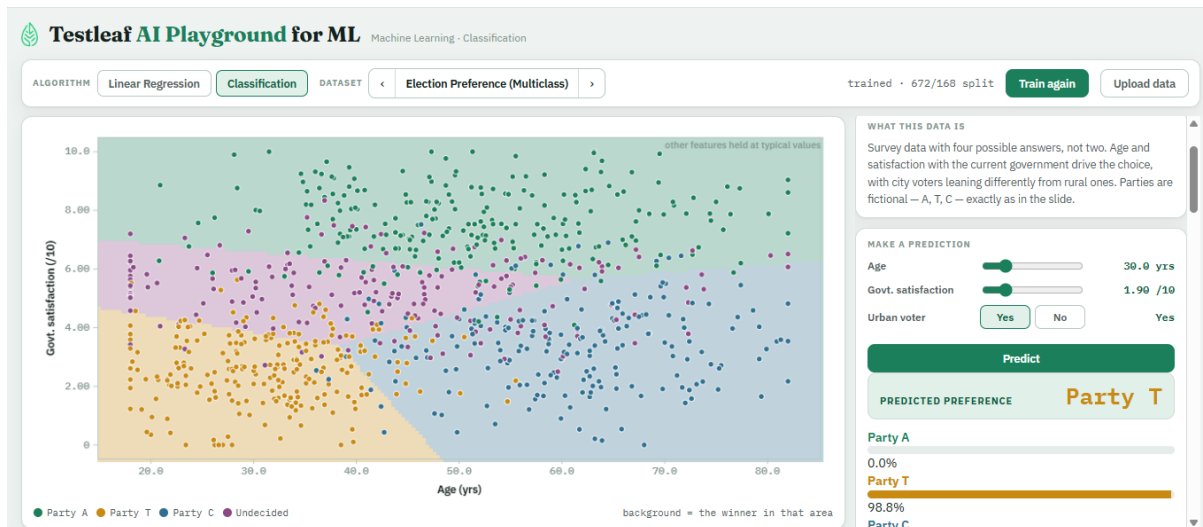


Food delivery at varying rates as per traffic and distance



1) Classification Algorithm

a) Age <30 and Govt satisfaction less ,urban



Party A -11.3 pts $\leftarrow$ base points + 0.031×30.0 (Age) = $+9.30 + 1.90 \times 1.90$ (Govt. satisfaction) = $+3.61 - 0.330 \times \text{Yes}$ (Urban voter) = $-0.330 \rightarrow -11.3 + 3.6 + 3.61 - .33 = 1.28$

Party T 11.4 pts $\leftarrow$ base points - 0.120×30.0 (Age) = $-3.6 - 1.63 \times 1.90$ (Govt. satisfaction) = $-3.10 + 1.83 \times \text{Yes}$ (Urban voter) = $+1.83$ ————— = **6.53 pts**

Party C 0.694 pts $\leftarrow$ base points + 0.107×30.0 (Age) = $+3.21 - 1.05 \times 1.90$ (Govt. satisfaction) = $-1.99 - 0.608 \times \text{Yes}$ (Urban voter) = -0.608 ————— = **1.30 pts**

Undecided 1.21 pts $\leftarrow$ base points - 0.017×30.0 (Age) = $-0.51 + 0.215 \times 1.90$ (Govt. satisfaction) = $+0.409 + 0.342 \times \text{Yes}$ (Urban voter) = $+0.342$ ————— = **1.4505 pts**

————— **highest score wins $\rightarrow$ Party T**

Party A $e^{1.28} = 3.59$

Party T $e^{6.53} = 685$

Party C $e^{1.3} = 3.67$

Undecided $e^{1.451} = 4.27$

Total (add all 4 up) = $696.59 \rightarrow 697$

Party A $3.59 \div 697 = 0.51\%$

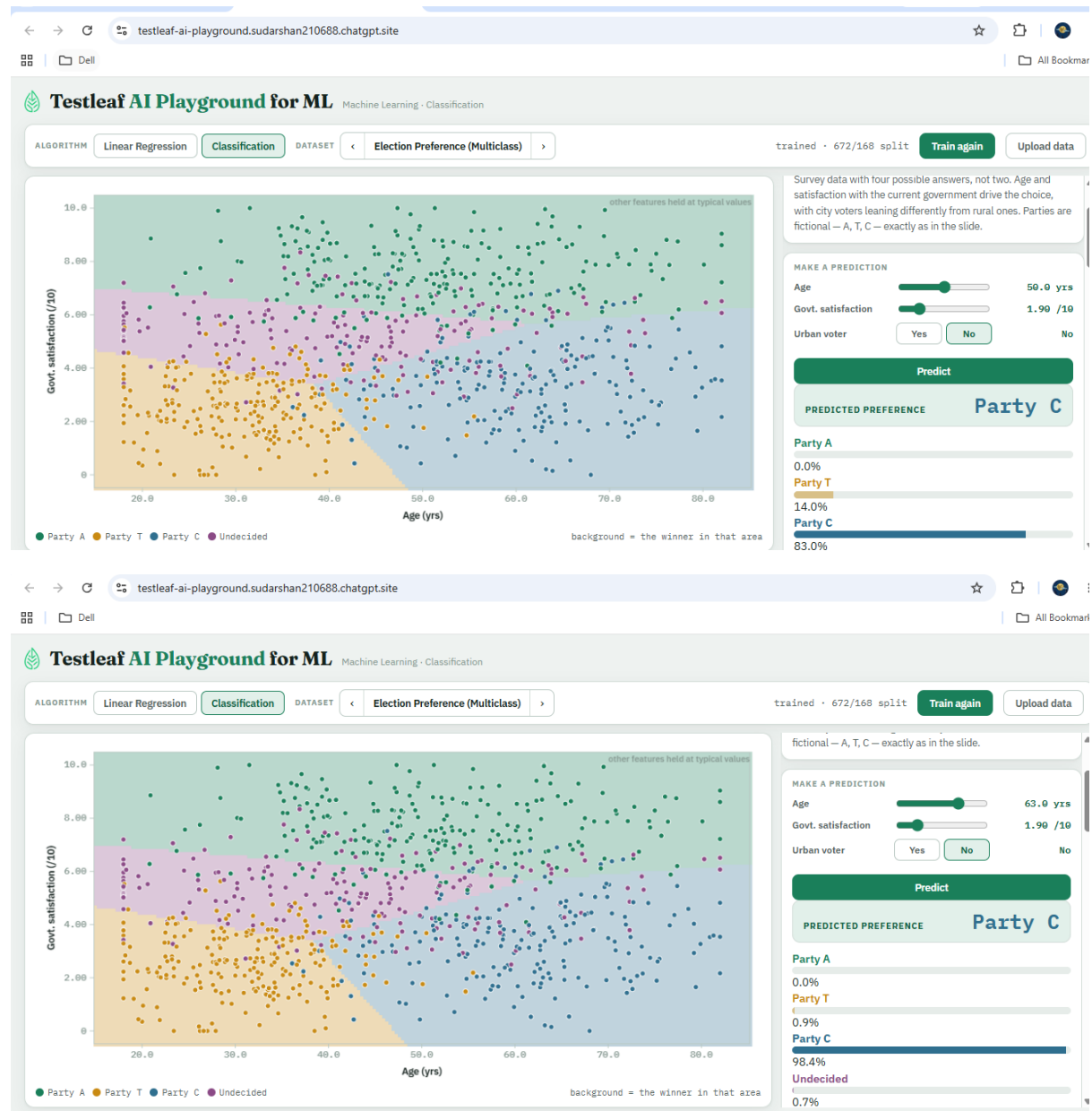
Party T $685 \div 697 = 98.28\%$

Party C $3.67 \div 697 = 0.52\%$

Undecided $4.27 \div 697 = 0.613\%$

b) Age >50 and Govt satisfaction less ,non urban

validate the percentage



Party A -11.3 pts $\leftarrow$ base points + 0.031×63.0 (Age) = +1.95 + 1.90×1.90 (Govt. satisfaction) = +3.61 – $0.330 \times \text{No}$ (Urban voter) = +0 ————— = -6.067 pts

Party T 11.4 pts $\leftarrow$ base points – 0.120×63.0 (Age) = -7.56 – 1.63×1.90 (Govt. satisfaction) = -3.10 + $1.83 \times \text{No}$ (Urban voter) = +0 ————— = 0.74 pts

Party C 0.694 pts $\leftarrow$ base points + 0.107×63.0 (Age) = +6.74 – 1.05×1.90 (Govt. satisfaction) = -1.995– $0.608 \times \text{No}$ (Urban voter) = +0 ————— = 5.439 pts

Undecided 1.21 pts $\leftarrow$ base points $- 0.017 \times 63.0$ (Age) $= -1.07 + 0.215 \times 1.90$ (Govt. satisfaction) $= +0.409 + 0.342 \times \text{No}$ (Urban voter) $= +0$ ————— **$= 0.549$ pts**
————— **highest score wins $\rightarrow$ Party C**

Party A $e^{-6.067} = 0.002$

Party T $e^{0.74} = 2.096$

Party C $e^{5.439} = 230.21$

Undecided $e^{0.549} = 1.73$

————— **Total (add all 4 up) = 234**

Party A $0.002 \div 234 = 0.0\%$

Party T $2.096 \div 234 = 0.89\%$

Party C $230.21 \div 234 = 98.38\%$

Undecided $1.73 \div 234 = 0.73\%$